Efficiency and acceleration in the accumulation of children's early vocabulary

Michael C. Frank

Abstract

Children typically learn hundreds of words over the first few years of their life, in a process that begins slowly but quickly picks up speed. Yet children vary widely in their rate of word learning, and this variation has important practical and theoretical consequences. Prior models of word learning describe vocabulary growth as a process of accumulation of evidence over time. Here we show that the process is best characterized as *accelerating* accumulation, with children varying in both their overall learning efficiency (intercept) and their acceleration rate (slope). This model provides a lever for understanding different loci of variation. Applied to cross-linguistic data, it suggests that the female advantage in early vocabulary results primarily from differences in overall learning efficiency, while socioeconomic differences appear in both terms and may compound through effects on acceleration rate. Using dense observations of children's early language environment, the model shows compounding effects on acceleration. In contrast, the effects of processing speed load primarily on efficiency and do not appear to compound. In contrast to children, large language models – even those trained on child-directed speech – are not well-described by this model: they show standard accumulator dynamics with no acceleration and very limited variability. In sum, our model provides a quantitative characterization of key signatures of early vocabulary growth, providing a target for future mechanistic learning models.

Children's early vocabulary grows very rapidly during early childhood, with large variation between children (Fenson et al. 1994, 2007; Frank et al. 2021). The growth has theoretical, practical, and methodological importance. From a theoretical perspective, growth mechanisms shed light on children's fundamental learning mechanisms and how they change across childhood (McMurray 2007; Bloom 2002; Frank et al. 2021). From a practical perspective, early vocabulary is an important precursor – and predictor – of academic achievement (Marchman and Fernald 2008; Catts et al. 2002; Muter et al. 2004). From a methodological perspective, vocabulary also affords a unique psychometric opportunity to study human variation because, unlike nearly all important human psychological metrics, there is a true zero point – a time before which children know no words. Thus, absolute levels of variability can be calculated.

One dominant viewpoint is that children's vocabulary growth can be characterized as a process of accumulation (McMurray 2007; Mitchell and McMurray 2009; Hidaka 2013; Mollica and Piantadosi 2017; Kachergis, Marchman, and Frank 2022). Each distinct word (type) can be conceptualized as a "bucket," and every token of language input then is a "drop" that falls into its respective bucket. When each bucket is full, the word is learned (McMurray 2007; Kachergis, Marchman, and Frank 2022). These models have been used to recover interpretable differences in "bucket size," often based on word properties including concreteness and syntactic category (Hidaka 2013; Kachergis, Marchman, and Frank 2022). They also provide a graded explanation for the widely-observed "vocabulary spurt" phenomenon – in which children's rate of vocabulary learning appears to increase dramatically (Frank et al. 2021; Ganger and Brent 2004; Gómez Díaz et al. 2024). While there are many other computational models of early word learning, in practice they have been more difficult to connect quantitatively with large-scale patterns of vocabulary growth (Frank, Goodman, and Tenenbaum 2009; McMurray, Horst, and Samuelson 2012; **trueswell2014?**; Kachergis, Yu, and Shiffrin 2012; Jones and Rowland 2017).

The accumulator view has been especially influential because it connects with a viewpoint in which the quantity of language that children hear plays a strong causal role in the speed of their learning (Fernald and Marchman 2012). Indeed, there is systematic evidence that variation in children's input – both its *quantity* and its *quality* – is related to variation in vocabulary (Hart and Risley 1995; Fernald and Marchman 2012; Weisleder and Fernald 2013; Anderson et al. 2021; Rowe 2012; Hirsh-Pasek et al. 2015; Coffey and Snedeker 2026). While input quantity is relatively easy to measure, measuring what makes input high quality has been challenging. One influential framework suggests that high quality input is interactionally rich, linguistically appropriate for the level of the child, and conceptually complex (Rowe and Snow 2020).

A naïve interpretation of the accumulator view would suggest that substantial variance in vocabulary growth should be related to input – either via quantity (number of drops) or quality (size of each drop). Two recent meta-analyses complicate this view, however. Anderson et al. (2021) showed that only a modest proportion of the variance in vocabulary outcomes is related to input quantity ($r^2 = .11$) and quality ($r^2 = 04$). A newer meta-analysis of quantity confirms and extends the proportion of variance due to quantity to 4–7% of total variation in vocabulary (Coffey and Snedeker 2026). Thus – as we might have supposed based on other research on human variation – not all variation in vocabulary between children can be attributed to input differences.

Further, work on children's word recognition suggests that children become faster and more accurate at matching words to their referent across early childhood (Bergelson 2020; Fernald et al. 1998; Fernald, Perfors, and Marchman 2006; Peter et al. 2019; Frank et al. 2026). Much of this work relies on the "looking while listening" (LWL) paradigm, in which children see two pictures (e.g. a book and a car) and their eye movements are recorded as they hear a sentence that refers to one of the pictures (e.g., "look at the book"). Children's reaction time to shift their eyes to the target (the book) provides an index of their speed of word recognition that decreases logarithmically across early childhood (Fernald et al. 2008; Frank et al. 2026) and relates to their vocabulary and other language outcomes (Fernald, Perfors, and Marchman 2006; Marchman and Fernald 2008; Fernald and Marchman 2012; Fernald, Marchman, and Weisleder 2013; Peter et al. 2019; Frank et al. 2026). One hypothesis proposed in this work is that greater language input leads to more rapid processing, in a virtuous cycle in which faster processing in turn leads to greater word learning (Weisleder and Fernald 2013). In the accumulator metaphor, one way to describe this proposal would be that individual children's buckets can be bigger or smaller based on their prior input and their processing efficiency, but to our knowledge this suggestion has not been fleshed out.

As this discussion indicates, despite the promise of accumulator models, they have largely not been used to understand variation between individuals. Instead, the majority of applications have focused on variation between words (Hidaka 2013; Kachergis, Marchman, and Frank 2022). Further, the rapidity of children's early word learning stands as a rebuke to a naive accumulator view. Sometimes exposure to a small number of examples of a word in the lab is enough for a child to form a lasting (Markson and Bloom 1997; Woodward, Markman, and Fitzsimmons 1994). In some cases, it seems only a small number of high-quality instances is sufficient for learning (Trueswell et al. 2013; Mollica and Piantadosi 2017), but if so, why do some words take much longer to learn?

The goal of the current work is to address these puzzles through systematic investigation and extension of accumulator models. All accumulator models seek to explain the same pattern: that the overall size of children's vocabulary increases supra-linearly. Where they differ is in what this growth means for the underlying learning process. Following previous work, we start with the idea that children's *internal learning ability* (not just their observed supra-linear vocabulary growth) accelerates across development (Hidaka 2013; Mayor and Plunkett 2010). We then extend a prior psychometric accumulator model (Kachergis, Marchman, and Frank 2022): our key addition is to describe variation between individual children by modeling both their individual learning efficiency and the acceleration of their learning. Following the bucket metaphor, this modification allows children to have different size buckets. This extension defines a longitudinal growth model that can be fit directly to large-scale datasets about children's actual vocabularies.

To constrain our model, we use data from the MacArthur-Bates Communicative Development Inventories [MB-CDI; Fenson et al. (1994);Marchman, Dale, and Fenson (2021)], a popular parent report instrument in which parents are asked to indicate which words their child produces and understands. Here we focus exclusively on production, which is used with older children and which tends to be a more reliable measure (Frank et al. 2021). We make use of longitudinal data from Wordbank (Frank et al. 2017), an open repository of data from the MB-CDI.

To address questions about the relationship between language input, language processing, and growth, we fit a Bayesian version of our growth model. We make use of new data resources to estimate the magnitude of linkages between input and vocabulary growth. First, we estimate plausible population values based on independent estimates of population variation in language input (Coffey et al. 2024). In addition, we use measures of individual children's language input from egocentric videos (Long et al. 2024) and day-long home recordings (Egan-Dailey and Bergelson 2025; Adams et al. 2018; Fernald, Marchman, and Weisleder 2013) to estimate the input-uptake relationship more precisely for a subsample of children, and measures of LWL word recognition from Peekbank (Zettersten et al. 2023), a large scale repository of LWL data.

Overall, our analysis supports a view in which children's vocabulary growth is not a pure process of accumulation but rather one that accelerates rapidly over time. We end by comparing the fit of our models on human data to their fit on large-language models. Making use of GPT-2 models trained on human developmental data, we show that – in contrast to children – these models *do* appear to function as pure accumulators. This analysis exposes a critical disanalogy between LLMs and human learners and suggests a potential route for understanding the vast "data gap" between children and large language models (Frank 2023; Warstadt et al. 2023).

Model

To begin our analysis, we present a nested set of accumulator-type models, many of which have been presented in the literature (see SI: Comparison with Prior Models). As we show below, each refinement of the model contributes to its ability to describe the variability between children. We start with the Rasch model from psychometrics (Rasch 1960; Bond and Fox 2013), which assumes

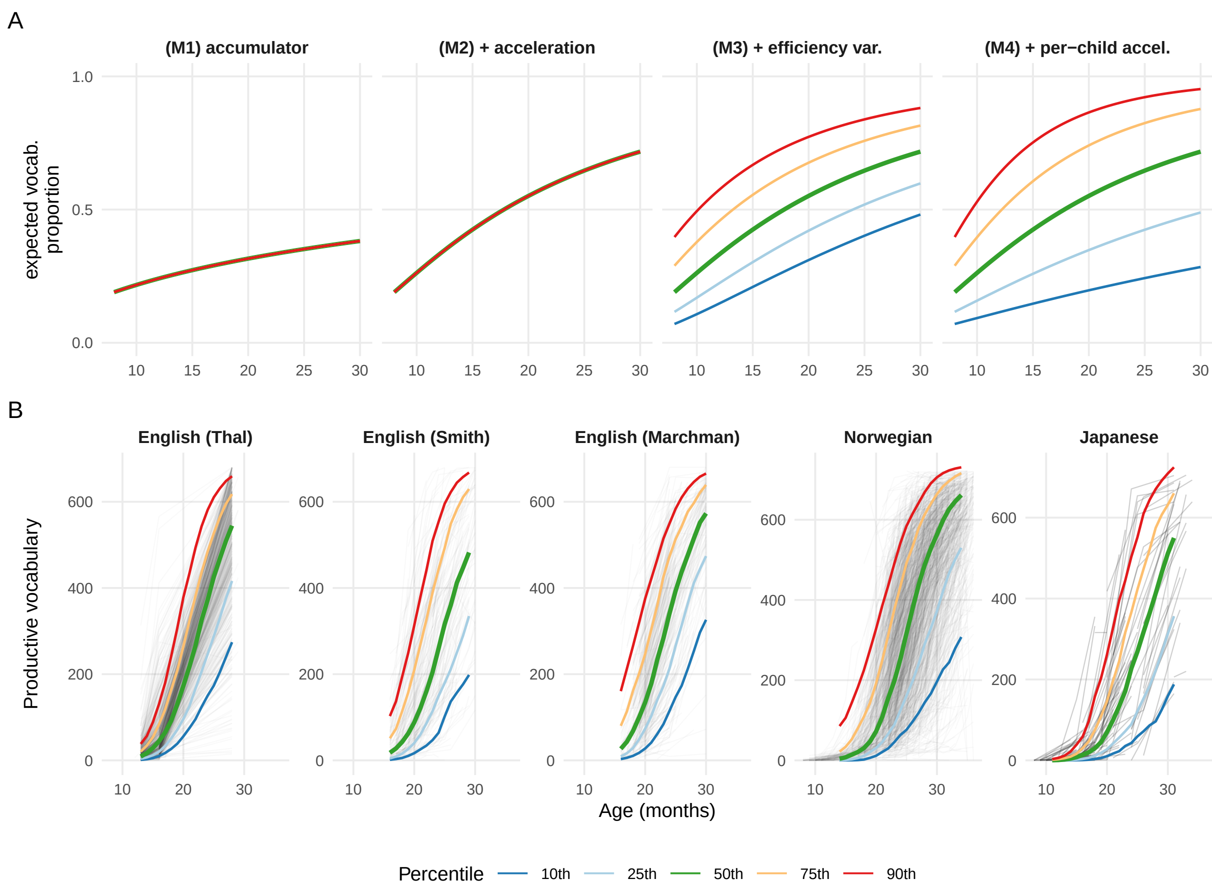

Figure 1: (A) Schematic predictions of all four models for vocabulary growth across age. Pure accumulator (M1) and accelerating accumulator (M2) models predict no variability across children, while M3 predicts variation and M4 predicts an increasing spread across chidren. (B) Best fitting model (M4) for five datasets. Gray lines show individual children's trajectories; colored lines show bootstrapped predictions at standard quantiles. Transparency is higher for panels with denser data to avoid overplotting.

that each child i has a learning ability θ_i and each word j has a difficulty δ_j.[1] The probability that child i produces word j is

$$P(w_{i,j} = 1 \mid \theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1 + \exp(\theta_i - \delta_j)}$$

Following Kachergis, Marchman, and Frank (2022), we can write our first model as a Rasch-style accumulator model (Model 1), specifying that θ_i is not a single parameter but instead a function of accumulation over time. In the simplest form:

$$\text{M1: } \theta_i(t) = \log(r_i) + \log(t/a_0)$$

where r_i is the child's rate of language input (in tokens per month) and t is the child's age (in months, anchored at a_0); note that we use log age to simulate linear accumulation because $\exp(\theta) = r \cdot t$. This form instantiates the basic hypothesis that the only thing that changes over time is more input, and the only source of variation between children is variation in rate.

We compare this model to an *accelerating accumulator* model (Model 2), in which the amount learned from each exposure increases with time by an exponent κ.

$$\text{M2: } \theta_i(t) = \log(r_i) + \kappa \ \log(t/a_0)$$

When $\kappa > 1$, each successive month yields more accumulation than the last.

Neither of these models has any way to capture variation across children beyond variation in input rate. Thus, we augment the basic accelerating accumulator, rewriting it as the sum of an individually variable efficiency term:

$$\text{M3: } \theta_i(t) = \xi_i + \kappa \ \log(t/a_0)$$

The intercept term $\xi_i = \log(r_i) + \log(\alpha_i)$ can then be used to break down individual variation into both input rate-related and other variation in children's efficiency.

Our final model includes individually variable efficiency and acceleration terms:

$$\text{M4: } \theta_i(t) = \xi_i + \kappa_i \ \log(t/a_0)$$

In this model, the slope term $\kappa_i = \kappa + \gamma \log r_i + \zeta_i$ adds *individual variation in acceleration*, decomposing acceleration into a population term, input-related acceleration, and a child-specific

[1] A variety of previous work has focused on estimating the properties of individual words (Goodman, Dale, and Li 2008; Hidaka 2013; Roy et al. 2015; Braginsky et al. 2019; Kachergis, Marchman, and Frank 2022; Kachergis et al. 2022). These analyses reveal that there is predictable word-level difficulty information but that only a modest portion of the variation between words is carried by frequency information. Words can be hard for a child to acquire because they vary also in the complexity and concreteness of their meanings, their syntactic category and syntactic context, their phonology, and many other factors. In the current work, we estimate word difficulty as a free parameter, and focus instead on variation between children.

residual acceleration ζ_i (the part not explained by input). The coefficient γ then captures the strength of the relationship between input and acceleration.

Model 4 instantiates the idea that children vary in their base rate of efficiency and their acceleration, creating variant of a standard longitudinal growth model in which children vary on their slope and intercept (Grimm, Ram, and Estabrook 2016). Note that, while we define the relationship of ξ_i and κ_i to a child's input r_i, we can nevertheless fit the simplest form of the model by just allowing these parameters to vary as random child-level slopes and intercepts.

As we will show, Model 4 provides a good description of variation in children's longitudinal growth trajectories. Figure 1 (top) shows schematic predictions from a pure accumulator model (M1), an accelerating accumulator (M2) in comparison to models including child-level variation in efficiency (M3) and both efficiency and acceleration (M4). Each model's predictions are represented as ability scores (θ) across age and then also projected into proportion correct scores on a discrete set of vocabulary items, representing an observed accuracy score that is subject to compression at floor and ceiling.

Our analysis generally depends on the availability of longitudinal data at the item level. The different models described above are not distinguishable from the cross-sectional, aggregate data that are typically considered in many analyses of early vocabulary. First, aggregate data do not allow for the estimation of individual word difficulties, thus the distribution of difficulties remains latent and different difficulty distributions can be posited to explain the accelerating shape of vocabulary growth (McMurray 2007; Mitchell and McMurray 2009). Second, cross-sectional data do not allow for identification of whether individual children's variation is due to differences in efficiency (intercept) or acceleration (slope) (see SI: Nonidentifiability in Cross-Sectional Data).

Results

Children's longitudinal growth reflects differences in both efficiency and acceleration

We first retrieved longitudinal data on monolingual, typically-developing children's vocabulary from Wordbank (Frank et al. 2017), an open archive of data from the MacArthur-Bates Communicative Development Inventory and its international adaptations. SI: Datasets shows the datasets used in each study in the paper; in this first analyses, we made use of five longitudinal datasets from Wordbank (three from American English and one each from Norwegian and Japanese, $N_{total} = 3135$).

Taking advantage of the equivalence of Rasch models to generalized linear mixed effects models (De Boeck et al. 2011), we fit the set of models shown in Figure 1 as a series of multi-level logistic regressions. Each model predicted the probability of an individual child producing a particular word as a function of age (either unmodified in M1 or multiplied by a constant in M2) and a random effect of word difficulty. Successive models add random intercepts (M3) and slopes (M4) by child. This initial set of fits simply attempts to describe the distribution of vocabulary growth curves.

Model 4, which incorporated variation in acceleration and efficiency, was the best fitting model across all datasets with differences in AIC > 1000 in all cases (see SI: AIC and BIC for Longitudinal Models). Figure 1 (bottom) shows the fit of the four model variants; the increasing spread of children's trajectories with age is especially noticeable for low-vocabulary older children [who may be classified as "late talkers"; Fernald and Marchman (2012); Rescorla (2011)].[2] Notably, variation in both efficiency and acceleration was unimodal across all samples (see SI: Characterizing Variation in Efficiency and Acceleration), providing no evidence of discrete clusters corresponding to language impairment (Pinstock and Schramm 2026; Singletary et al. 2025).

Efficiency and acceleration relate to separate demographic predictors

Children's vocabulary is known to be related to a number of sociodemographic predictors, including sex and maternal education (which is interpreted as a proxy for socioeconomic status). In particular, female children show consistently larger vocabularies than male children across dozens of languages and countries (Eriksson et al. 2012; Frank et al. 2021). Similarly, children of more educated mothers tend to show larger vocabularies, though this effect is somewhat more heterogeneous across countries and languages (Frank et al. 2021).

Our model offers a summary of children's growth in terms of both their efficiency (intercept) and their acceleration (slope). This decomposition allows us to ask whether these two demographics effects – sex and maternal education – are seen in efficiency, acceleration, or both. For each child in our analysis, we extracted the best linear unbiased prediction (BLUP) from our models. We then regressed intercept and slope terms against each demographic predictor separately for each language. Although we cannot fit individual children's growth trajectories in cross-sectional data, we can also in principle apply the same model to recover population parameters based on cross-sectional data, so we did so for an additional group of datasets for which sex and maternal education information were available.

Figure 2 shows the per-language coefficients from this analysis, along with a cross-language meta-analytic estimate for both the cross-sectional and longitudinal data (which largely agree). Examining the cross-sectional data since this analysis includes more datasets, the female advantage is apparent across all datasets, but this effect manifests itself more substantially efficiency ($\beta = 0.54$, $p =< .001$) than acceleration ($\beta = 0.35$, $p =< .001$). In contrast, maternal education effects were present in both measures, but stronger in acceleration (efficiency: $\beta = 0.1$, $p = .004$; acceleration: $\beta = 0.38$, $p =< .001$). One interpretation of these findings is that sex effects reflect a small constant offset in average ability. In contrast, socioeconomic differences – as reflected in maternal education – might continue to compound across development (Fernald, Marchman, and Weisleder 2013; Hart and Risley 2003).

Population input-related variation corresponds to meta-analytic estimates

In our prior model fits, we fit population variation in efficiency and acceleration without considering the connection to language input. We next consider whether this relationship can be estimated at a population level. We use markov chain monte carlo to fit a version of model 4 with a linkage between efficiency and input rate such that $\xi_i = \log(r_i) + \log(\alpha_i)$, where r_i is not known but we assume that there is a population variation in language input rate: σ_r. This model – which does not include individual child-level input data – can not distinguish whether input contributes to efficiency

[2] Following the literature, we primarily investigate linear accumulator models, which show power-law growth of ability over time because θ is a function of $\log(t)$. However, it is also possible that ability grows linearly in time, which would lead to exponential increases in the probability of word production. We fit linear-time models, but found that they were dispreferred statistically in all but one case, though differences were relatively small within the restricted age range for which we have data (See SI: Full Cross-linguistic Model Ladder).

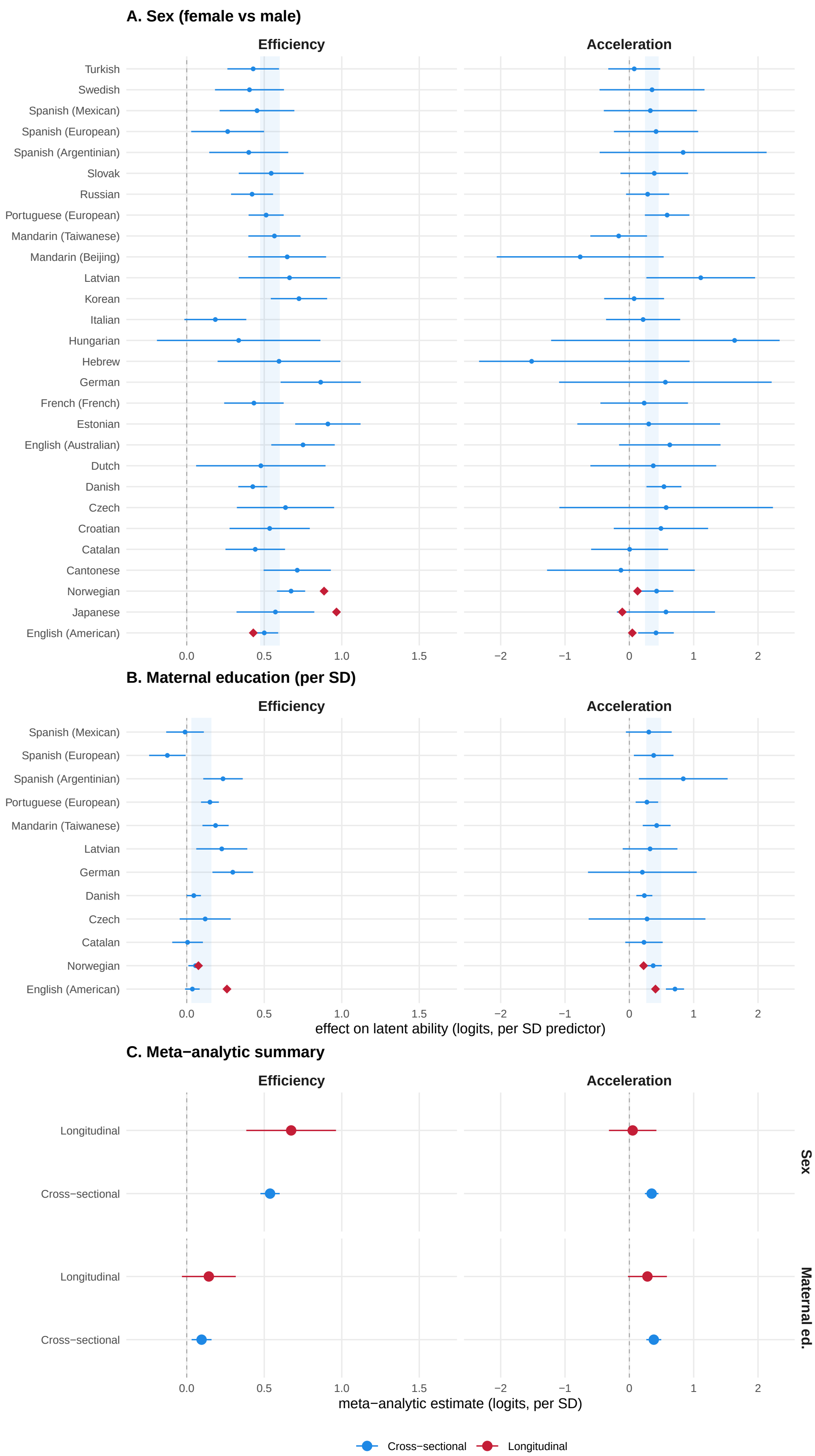

Figure 2: Regression coefficients predicting individual children's efficiency and acceleration as a function of gender and maternal education. Panels A and B show data from individual languages (both longitudinal and cross-sectional); blue regions show meta-analytic estimates from a random-effects meta analysis. Error bars show 95% confidence intervals. Panel C shows a meta-analytic summary for cross-sectional and longitudinal datasets.

or instead to acceleration. Instead, because the accumulator model makes predictions about links between the actual amount of input children receive and what they learn, this model can ask what proportion of observed variation between children is plausibly due to input-related variation.

Fitting this model requires a guess about the range of input variation present in the populations for which we have dense longitudinal data (English-learning children in the US and Norwegian-learning children in Norway). Leaving aside the question of the specific sampling processes that led to our data, estimating children's input range is notoriously tricky: The rate of language that children hear varies across cultures, families, activities, and measurement occasions (Coffey et al. 2024). Thus, we consider the share of variation due to input across a range of plausible values, anchored via a range of estimates in the published literature (see SI: Population Input Range Estimation).

Figure 3 (panel F) shows the results models fit with three values for each language, interpolated with an analytic estimate of the relationship between σ_r and the total proportion of variance in efficiency that could be attributed to input. Within the range of plausible input values from the literature, we observed a strong congruence between the model estimates of input variance (EN: 3.5–9.1%, NO: 1.8–4.6%) and the meta-analytic range reported by Coffey and Snedeker (2026) (4–7%). This consistency both supports the overall plausibility of the model and provides a second route for estimating the significant but modest contribution of input quantity to variation in children's outcomes.

Input predicts both efficiency and acceleration, while processing speed relates only to efficiency

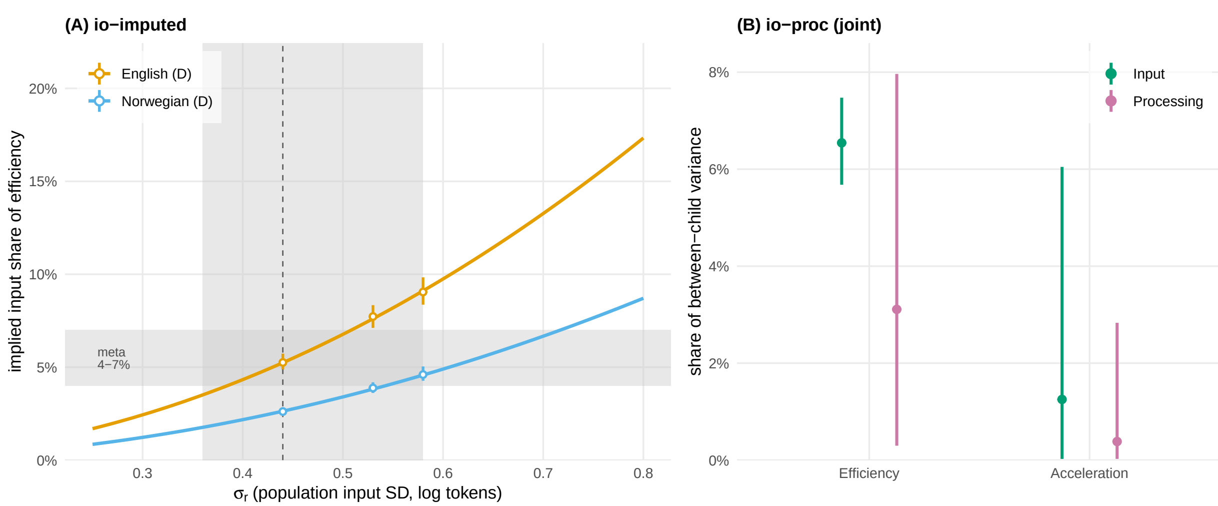

Figure 3

Population estimates of input sensitivity do not offer a direct estimate of input couplings within individual children, and they do not allow us to estimate the separate contributions of language input and language processing to efficiency and acceleration.For this, we turn to a set of five longitudinal datasets (described shown in SI: Datasets) that include either dense home recordings, longitudinal LWL processing data, or both. Three include recordings from day-long audio recordings using LENA (Language Environment Analysis) recorders, which offer an automated adult word count estimate that has been validated by transcription studies (Cristia et al. 2021). One includes short video recordings of egocentric video recorded in children's homes, with word counts estimated from automated transcripts (Long et al. 2024). Each of these metrics provides a noisy proxy for total home input, but nevertheless allows us to estimate models that include individual input. Further, four of the five include multiple LWL sessions per individual, offering data on children's word recognition speed (w_i).

We investigated several variants of Model 4 (See SI: Predicting a version of Model 4 to this dataset, with modifications to allow input and acceleration to contribute to both acceleration and efficiency. We then fit this model to the joint dataset but with observations of r_i and w_i for individual children.

Figure 3 (B) shows small but highly reliable effects of language input on both efficiency and acceleration. These effects are also in the range of the meta-analyses, but they suggest that input-related differences – though small – could compound through their effect on acceleration. In contrast, processing predicted *only* efficiency and not acceleration.

Growth in ability leads to rapid changes in learning times for individual words

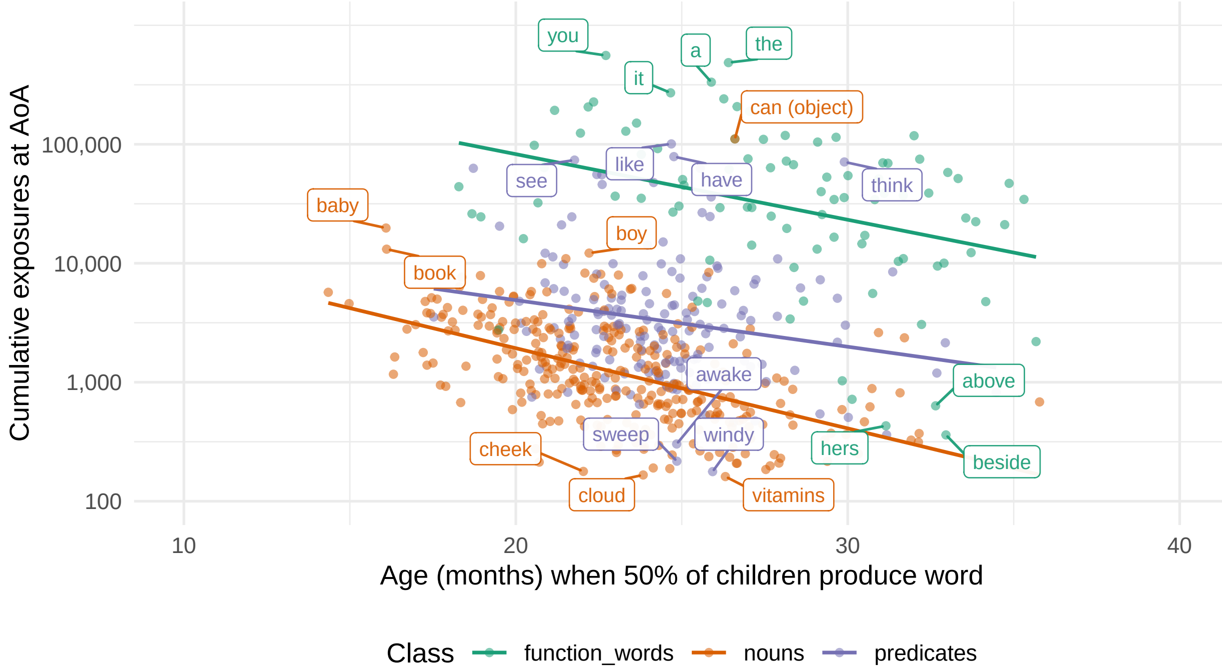

Figure 4

As an illustration of the changes in efficiency and acceleration that we observe in our data, we computed the average proportion of exposures necessary for a child to learn a word across ages. Figure 4 shows a model-derived estimate of the average number of times a word is heard before more than 50% of children produce it (See SI: Changes in Word Efficiency), using frequency estimates from the CHILDES Corpus (MacWhinney 2000; Sanchez et al. 2019).

Across different word classes (e.g., nouns, verbs, closed class words), efficiency increases exponentially, such that early-learned words such as "book" may be heard tens of thousands of times before they are produced, while later learned words such as “vitamin” might only be heard hundreds times before being produced. Indeed, this analysis likely understates the degree of change in efficiency as it assumes that each exposure to a word is equally meaningful. In fact, individual exposures vary widely in how much information they give. Sometimes words are overheard rather than directed to the child, or lack physical context to allow for clear inference (Cartmill et al. 2013; Trueswell et al. 2013; Mollica and Piantadosi 2017). Thus, in line with both intuition and experiments, the total number of exposures required on average to learn a word might indeed be only a small handful by age three (Markson and Bloom 1997; Bloom 2002).

Large language models show no acceleration in their vocabulary growth

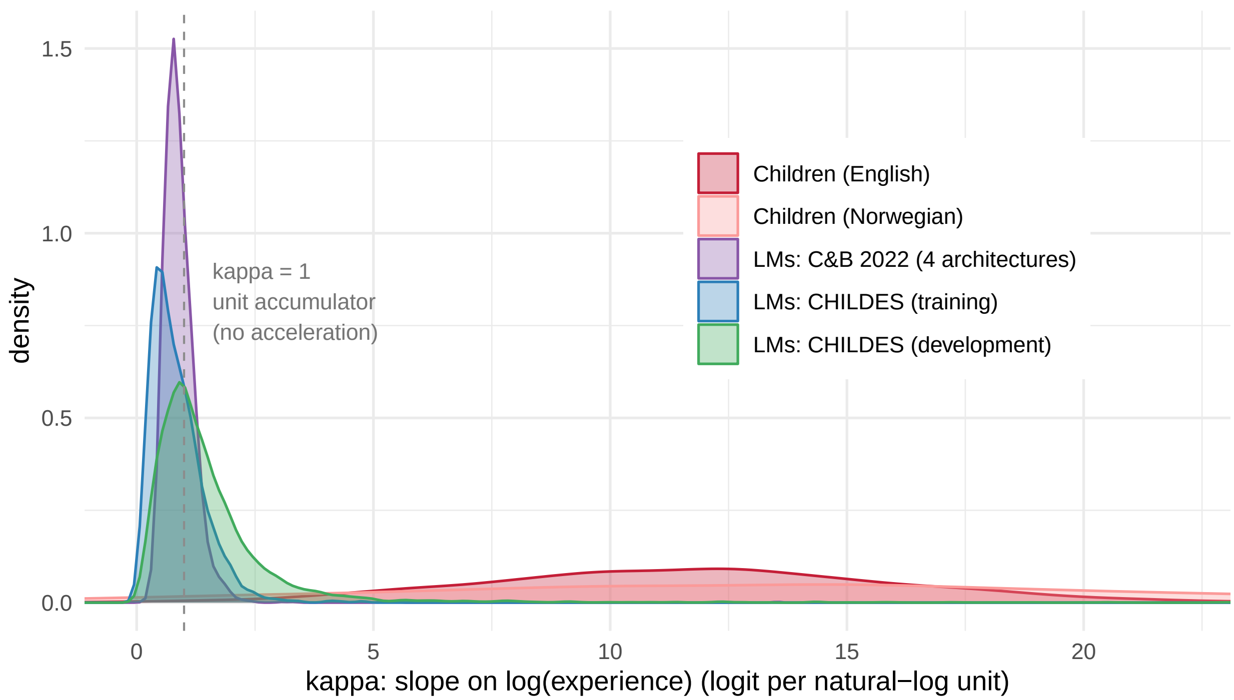

Figure 5: The density of acceleration slopes for language models (by word) and children (by child), extracted from English and Norwegian models.

Small-scale language models trained on developmentally-realistic corpora have become a useful tool for understanding early language development (Huebner et al. 2021; Futrell and Mahowald 2025; Frank and Goodman 2025). One major disanalogy between humans and language models, however, is the massive scale of training data necessary for language models to achieve high levels of performance (Warstadt et al. 2023; Frank 2023).

For our final experiments, we were interested in whether language models show the same patterns of developmental change in efficiency or acceleration. We make use of work by Chang and Bergen (2022), who extracted per-word acquisition curves from language models. They computed logistic fits to the surprisal (negative log probability) of individual words across training, estimating a logistic curve for each word, just as our Rasch model does. Critically for our purposes, the slope of these logistic curves provides a direct analogue to our κ parameter capturing acceleration in the accumulation process (see SI: Computing Acceleration in LMs).

We compared the word-by-word acceleration of small-scale LLMs to children across several experiments. First, we extracted slopes from fits provided by Chang and Bergen (2022), in which four different language model architectures (LSTM, BiLSTM, GPT-2, and BERT) were fit to standard corpora (BookCorpus and WikiText-103). This training setup was disanalogous with our studies of children in at least two ways, however. First, models were not trained on child directed speech, so we retrained new GPT-2 (small) models on 24.5M words from CHILDES, following the protocol in Feng, Goodman, and Frank (2024). Second, unlike children, both our first set of models and those trained by Chang and Bergen (2022) were trained to convergence, with acquisition curves reflecting changing predictions across repeated passes through the same training data. To isolate developmental change, we trained a separate set of GPT-2 models each to convergence on increasingly larger subsamples of the same CHILDES data, allowing us to estimate surprisal as a function of actual amount of input data – a more direct developmental analogue.

The results across all three sets of models were clear (Figure 5): κ for all models was clustered near 1, the unit accumulator posited by McMurray (2007). In contrast, κ estimates for children for our large longitudinal samples were much higher (EN: VALUE NOT SPECIFIED and NO: VALUE NOT SPECIFIED) and highly variable across children (EN SD: VALUE NOT SPECIFIED and NO SD: VALUE NOT SPECIFIED). In sum, language models – but not children – behave like Model 1 (an accumulator). In contrast, children are best described by Model 4 – they are variable, accelerating accumulators.

Discussion

How does children's vocabulary grow over early childhood? We proposed that this process is characterized as accelerating accumulation, in which variation between children is produced by differences in both the efficiency of the accumulation process and the rate of acceleration. This cognitive process leads to a simple two parameter longitudinal growth model that provides a good description of cross-linguistic data.

These two parameters provide a set of measures with which to distinguish a number of important factors that are related to overall vocabulary size in prior work. Factors that load on efficiency alone relate to children's overall learning trajectory, creating a constant offset in language ability. In contrast, those that load on acceleration create an increasing fan, such that older children are more different from one another than younger ones. On the side of demographics, we found evidence that while sex differences manifest as differences in efficiency, socioeconomic differences show up in acceleration as well. Evidence about language input paints a similar picture: while language processing measures such as LWL index differences in efficiency, language input variation appears to affect acceleration as well. Thus, our model identifies a channel by which experience might compound across development.

Where does the acceleration in children's learning come from? One possibility is that it is driven by developmental maturation of general abilities such as learning, memory, and attention – such changes could mean that children's "buckets" require fewer drops to fill them. Another possibility is that children's increasingly sophisticated inferential abilities allow them to extract progressively more information from the linguistic signal [Smith et al. (2002);Markman and Wachtel (1988);Bohn and Frank (2019);meylan2022]; Mitchell and McMurray (2009) study how these sorts of "leverage" mechanisms can change the slope of acceleration in accumulator models. Network growth models provide one promising avenue for understanding how leverage might play out in specific patterns of vocabulary growth at the level of word content (Hills et al. 2009; Fourtassi, Bian, and Frank 2020).

Our work here has a number of limitations that suggest avenues for new work. First, while we analyzed a range of cross-linguistic data where it was available, most of the open data on children's input and processing are available only in English, and for convenience samples. Thus, the specific numerical estimates that we were able to derive are at best approximations and do not represent estimates for broader populations of interest. Systematic representative sampling would be required for generalization. Further, the studies we included here focused exclusively on monolingual, typically-developing children. Though we have reason to expect that these mechanisms would apply in multilingual acquisition, significant work will be necessary to adapt and test our proposal in multilingual datasets (Gómez Díaz et al. 2024; Tan, Marchman, and Frank 2024).

In addition, we focused on the role of input quantity as a key first-order factor influencing the accumulation of language. However, a rich tradition describes quantity-related variation as well (e.g., Anderson et al. 2021; Rowe 2012; Hirsh-Pasek et al. 2015). While individual studies have identified a wide variety of indicators of quality, they are still challenging to compute at scale from audio and video recordings. Extracting these and connecting them to growth over time is an exciting prospect for future work.

Finally, our study is a computational analysis of observational data; it provides at best hypotheses about causal structure. Our hope is that by providing signatures of longitudinal variation, we can inspire further work using both dense measurement and intervention that can gain causal traction on the mechanisms of vocabulary growth.

Two often, psychology and cognitive science focus either only on universal mechanisms or variability between individuals. Word learning is a unique case study in cognitive science because it describes a basic process of knowledge acquisition that also captures large, meaningful individual variation. As such, it becomes possible to link mechanistic models of learning (such as accumulators) to the signatures that form the locus of variation in human learners.

The accumulator idea that forms the basis of our model is extremely simple, but it is potentially compatible with a wide range of more elaborated computational models of early language learning (Frank, Goodman, and Tenenbaum 2009; McMurray, Horst, and Samuelson 2012; **trueswell2014?**; Kachergis, Yu, and Shiffrin 2012). One important direction for future work is to survey and analyze models from the literature to understand which of these show learning dynamics congruent with the statistical signatures we have identified in actual children's vocabulary growth. We have shown that one class of models does not show these signatures, however.

Modern language models (both transformer-based and recurrent) failed to manifest the acceleration that we found pervasively in our data. This lack of "leverage" in such models is one potential clue to why they require such massive amounts of data to learn, relative to human children.

Methods

Data and Code

Data were retrieved from Wordbank using the `wordbankr` package (Frank et al. 2017), childes-db VER using `childesr` (Sanchez et al. 2019), and peekbank version 2026.1 using `peekbankr`

(Zettersten et al. 2023). BabyView data are from release 2025.2, specifically those used by (**tan2026?**). SEEDLings data are downloaded from the materials provided by Egan-Dailey and Bergelson (2025).

Reproducible code for this paper is available at http://github.com/langcog/standard-model-2.

Models

Frequentist models

Initial models were fit with `glmer` (Bates et al. 2015), with the following specifications:

- M1: `produces ~ offset(log_age) + (1 | item)`
- M2: `produces ~ 1 + log_age + (1 | item)`
- M3: `produces ~ 1 + log_age + (1 | item) + (1 | child)`
- M4: `produces ~ 1 + log_age + (1 | item) + (log_age | child)`

Linear variants reported in the supplement used linear age rather than log age.

Bayesian models

Bayesian models were fit using Stan (Carpenter et al. 2017).

Priors

Inference

Input and processing models

Input

GPT-2 models

We trained XYZ

Acknowledgements

Some of the computing for this project was performed on the Sherlock and Marlowe clusters. We would like to thank Stanford University and Stanford Research Computing for providing computational resources and support that contributed to these research results. Funds for computational experiments were also provided by a gift from Meta to support research on children's home language environment. Claude Code was used for literature review, code generation, visualization, and simulation infrastructure.

References

Adams, Katherine A., Virginia A. Marchman, Elizabeth C. Loi, Melanie D. Ashland, Anne Fernald, and Heidi M. Feldman. 2018. "Caregiver Talk and Medical Risk as Predictors of Language Outcomes in Full Term and Preterm Toddlers." *Child Development* 89 (5): 1674–90.

Anderson, Nina J., Susan A. Graham, Heather Prime, Jennifer M. Jenkins, and Sheri Madigan. 2021. "Linking Quality and Quantity of Parental Linguistic Input to Child Language Skills: A Meta-Analysis." *Child Development* 92 (2): 484–501.

Bates, Douglas, Martin Mächler, Ben Bolker, and Steve Walker. 2015. "Fitting Linear Mixed-Effects Models Using lme4." *Journal of Statistical Software* 67 (1): 1–48. https://doi.org/10.18637/jss.v067.i01.

Bergelson, Elika. 2020. "The Comprehension Boost in Early Word Learning: Older Infants Are Better Learners." *Child Development Perspectives* 14 (3): 142–49.

Bloom, Paul. 2002. *How Children Learn the Meanings of Words.* MIT press.

Bohn, Manuel, and Michael C Frank. 2019. "The Pervasive Role of Pragmatics in Early Language." *Annual Review of Developmental Psychology* 1 (1): 223–49.

Bond, Trevor G, and Christine M Fox. 2013. *Applying the Rasch Model: Fundamental Measurement in the Human Sciences.* Psychology Press.

Braginsky, Mika, Daniel Yurovsky, Virginia A Marchman, and Michael C Frank. 2019. "Consistency and Variability in Children's Word Learning Across Languages." *Open Mind* 3: 52–67.

Carpenter, Bob, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus A. Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. 2017. "Stan: A Probabilistic Programming Language." *Journal of Statistical Software* 76 (1): 1–32. https://doi.org/10.18637/jss.v076.i01.

Cartmill, Erica A, Benjamin F Armstrong III, Lila R Gleitman, Susan Goldin-Meadow, Tamara N Medina, and John C Trueswell. 2013. "Quality of Early Parent Input Predicts Child Vocabulary 3 Years Later." *Proceedings of the National Academy of Sciences* 110 (28): 11278–83.

Catts, Hugh W, Marc E Fey, J Bruce Tomblin, and Xuyang Zhang. 2002. "A Longitudinal Investigation of Reading Outcomes in Children with Language Impairments." *Journal of Speech, Language, and Hearing Research* 45 (6): 1142–57.

Chang, Tyler A., and Benjamin K. Bergen. 2022. "Word Acquisition in Neural Language Models." *Transactions of the Association for Computational Linguistics* 10: 1–16. https://doi.org/10.1162/tacl_a_00444.

Coffey, Joseph R., Okko Räsänen, Camila Scaff, and Alejandrina Cristia. 2024. "The Difficulty and Importance of Estimating the Lower and Upper Bounds of Infant Speech Exposure." In *Proceedings of Interspeech 2024*, 3615. Kos, Greece. https://doi.org/10.21437/Interspeech.2024-2102.

Coffey, Joseph R., and Jesse Snedeker. 2026. "How Strong Is the Relationship Between Caregiver Speech and Language Development? A Meta-Analysis." *Journal of Child Language* 53: 435–70. https://doi.org/10.1017/S0305000924000692.

Cristia, Alejandrina, Marvin Lavechin, Camila Scaff, Melanie Soderstrom, Caroline Rowland, Okko Räsänen, John Bunce, and Elika Bergelson. 2021. "A Thorough Evaluation of the Language Environment Analysis (LENA) System." *Behavior Research Methods* 53 (2): 467–86.

De Boeck, Paul, Marjan Bakker, Robert Zwitser, Michel Nivard, Abe Hofman, Francis Tuerlinckx, and Ivailo Partchev. 2011. "The Estimation of Item Response Models with the Lmer Function from the Lme4 Package in r." *Journal of Statistical Software* 39: 1–28.

Egan-Dailey, Shannon, and Elika Bergelson. 2025. "Early Child Measures Outpredict Input Measures of Preschool Language Skills in US English Learners." *Developmental Psychology.*

Eriksson, Mårten, Peter B Marschik, Tiia Tulviste, Margareta Almgren, Miguel Pérez Pereira, Sonja Wehberg, Ljubica Marjanovič-Umek, Frederique Gayraud, Melita Kovacevic, and Carlos Gallego.

2012. “Differences Between Girls and Boys in Emerging Language Skills: Evidence from 10 Language Communities.” *British Journal of Developmental Psychology* 30 (2): 326–43.

Feng, Steven Y., Noah D. Goodman, and Michael C. Frank. 2024. “Is Child-Directed Speech Effective Training Data for Language Models?” In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, edited by Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, 22055–71. Miami, Florida, USA: Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.emnlp-main.1231.

Fenson, Larry et al. 2007. *MacArthur-Bates Communicative Development Inventories.* Baltimore, MD: Paul H. Brookes Publishing Company.

Fenson, Larry, Philip S Dale, J Steven Reznick, Elizabeth Bates, Donna J Thal, Stephen J Pethick, Michael Tomasello, Carolyn B Mervis, and Joan Stiles. 1994. “Variability in Early Communicative Development.” *Monographs of the Society for Research in Child Development*, i–185.

Fernald, Anne, and Virginia A Marchman. 2012. “Individual Differences in Lexical Processing at 18 Months Predict Vocabulary Growth in Typically Developing and Late-Talking Toddlers.” *Child Development* 83 (1): 203–22.

Fernald, Anne, Virginia A Marchman, and Adriana Weisleder. 2013. “SES Differences in Language Processing Skill and Vocabulary Are Evident at 18 Months.” *Developmental Science* 16 (2): 234–48.

Fernald, Anne, Amy Perfors, and Virginia A. Marchman. 2006. “Picking up Speed in Understanding: Speech Processing Efficiency and Vocabulary Growth Across the 2nd Year.” *Developmental Psychology* 42 (1): 98–116. https://doi.org/10.1037/0012-1649.42.1.98.

Fernald, Anne, John P. Pinto, Daniel Swingley, Amy Weinberg, and Gerald W. McRoberts. 1998. “Rapid Gains in Speed of Verbal Processing by Infants in the 2nd Year.” *Psychological Science* 9 (3): 228–31.

Fernald, Anne, Renate Zangl, Ana Luz Portillo, and Virginia A. Marchman. 2008. “Looking While Listening: Using Eye Movements to Monitor Spoken Language Comprehension by Infants and Young Children.” In *Developmental Psycholinguistics: On-Line Methods in Children’s Language Processing*, edited by Irina A. Sekerina, Eva M. Fernandez, and Harald Clahsen, 97–135. Amsterdam: John Benjamins.

Fourtassi, Abdellah, Yuan Bian, and Michael C Frank. 2020. “The Growth of Children’s Semantic and Phonological Networks: Insight from 10 Languages.” *Cognitive Science* 44 (7): e12847. https://doi.org/10.1111/cogs.12847.

Frank, Michael C. 2023. “Bridging the Data Gap Between Children and Large Language Models.” *Trends in Cognitive Sciences* 27 (11): 990–92.

Frank, Michael C, Mika Braginsky, Daniel Yurovsky, and Virginia A Marchman. 2021. *Variability and Consistency in Early Language Learning: The Wordbank Project.* MIT Press.

Frank, Michael C, Mika Braginsky, Daniel Yurovsky, and Virginia A. Marchman. 2017. “Wordbank: An Open Repository for Developmental Vocabulary Data.” *Journal of Child Language* 44 (3): 677–94.

Frank, Michael C, and Noah D Goodman. 2025. “Cognitive Modeling Using Artificial Intelligence.” *Annual Review of Psychology* 77.

Frank, Michael C, Noah D Goodman, and Joshua B Tenenbaum. 2009. “Using Speakers’ Referential Intentions to Model Early Cross-Situational Word Learning.” *Psychological Science* 20 (5): 578–85.

Frank, Michael C, Virginia A Marchman, Claire Augusta Bergey, Veronica Boyce, Mika Braginsky, George Kachergis, Jess Mankewitz, et al. 2026. "Continuous Developmental Changes in Word Recognition Support Language Learning Across Early Childhood." *eLife* 14.

Futrell, Richard, and Kyle Mahowald. 2025. "How Linguistics Learned to Stop Worrying and Love the Language Models." *Behavioral and Brain Sciences*, 1–98.

Ganger, Jennifer, and Michael R. Brent. 2004. "Reexamining the Vocabulary Spurt." *Developmental Psychology* 40 (4): 621–32. https://doi.org/10.1037/0012-1649.40.4.621.

Gómez Díaz, Miranda, Laia Fibla, Rachel Ka-Ying Tsui, and Krista Byers-Heinlein. 2024. "Testing Theories of the Vocabulary Spurt with Monolingual and Bilingual Infants." *Developmental Psychology* 60 (8): 1357–71. https://doi.org/10.1037/dev0001777.

Goodman, Judith C, Philip S Dale, and Ping Li. 2008. "Does Frequency Count? Parental Input and the Acquisition of Vocabulary." *Journal of Child Language* 35 (3): 515–31.

Grimm, Kevin J, Nilam Ram, and Ryne Estabrook. 2016. *Growth Modeling: Structural Equation and Multilevel Modeling Approaches*. Guilford Publications.

Hagihara, Hiromichi, Monica Barbir, Mikako Ishibashi, Yasuhiro Kanakogi, Masaharu Kato, Irena Lovcevic, Youtao Lu, et al. 2023. "A Sharable Merged Dataset on Japanese Children's Vocabulary Measures Using the Japanese MacArthur–Bates Communicative Development Inventory." OSF. https://doi.org/10.17605/osf.io/s5ydw.

Hart, Betty, and Todd R Risley. 1995. *Meaningful Differences in the Everyday Experience of Young American Children*. Brookes.

———. 2003. "The Early Catastrophe: The 30 Million Word Gap by Age 3." *American Educator* 27 (1): 4–9.

Hidaka, Shohei. 2013. "A Computational Model Associating Learning Process, Word Attributes, and Age of Acquisition." *PloS One* 8 (11).

Hills, Thomas T., Mounir Maouene, Josita Maouene, Adam Sheya, and Linda Smith. 2009. "Longitudinal Analysis of Early Semantic Networks: Preferential Attachment or Preferential Acquisition?" *Psychological Science* 20 (6): 729–39. https://doi.org/10.1111/j.1467-9280.2009.02365.x.

Hirsh-Pasek, Kathy, Lauren B Adamson, Roger Bakeman, Margaret Tresch Owen, Roberta Michnick Golinkoff, Amy Pace, Paula KS Yust, and Katharine Suma. 2015. "The Contribution of Early Communication Quality to Low-Income Children's Language Success." *Psychological Science* 26 (7): 1071–83.

Huebner, Philip A., Elior Sulem, Cynthia Fisher, and Dan Roth. 2021. "BabyBERTa: Learning More Grammar with Small-Scale Child-Directed Language." In *Proceedings of the 25th Conference on Computational Natural Language Learning (CoNLL)*, 624–46. https://doi.org/10.18653/v1/2021.conll-1.49.

Jones, Gary, and Caroline F Rowland. 2017. "Diversity Not Quantity in Caregiver Speech: Using Computational Modeling to Isolate the Effects of the Quantity and the Diversity of the Input on Vocabulary Growth." *Cognitive Psychology* 98: 1–21.

Kachergis, George, Virginia A Marchman, Philip S Dale, Jessica Mankewitz, and Michael C Frank. 2022. "Online Computerized Adaptive Tests of Children's Vocabulary Development in English and Mexican Spanish." *Journal of Speech, Language, and Hearing Research* 65 (6): 2288–308.

Kachergis, George, Virginia A Marchman, and Michael C Frank. 2022. "Toward a 'Standard Model' of Early Language Learning." *Current Directions in Psychological Science* 31 (1): 20–27.

Kachergis, George, Chen Yu, and Richard M Shiffrin. 2012. "An Associative Model of Adaptive

Inference for Learning Word–Referent Mappings." *Psychonomic Bulletin & Review* 19 (2): 317–24.

Long, Bria, Robert Z Sparks, Violet Xiang, Stefan Stojanov, Zi Yin, Grace E Keene, Alvin WM Tan, et al. 2024. "The BabyView Dataset: High-Resolution Egocentric Videos of Infants' and Young Children's Everyday Experiences." *arXiv Preprint arXiv:2406.10447*.

MacWhinney, Brian. 2000. *The CHILDES Project: Tools for Analyzing Talk*. Vol. 1. Psychology Press.

Marchman, Virginia A, Philip Dale, and Larry Fenson. 2021. *MacArthur-Bates Communicative Development Inventories User's Guide and Technical Manual, Third Edition*. Brookes Publishing.

Marchman, Virginia A, and Anne Fernald. 2008. "Speed of Word Recognition and Vocabulary Knowledge in Infancy Predict Cognitive and Language Outcomes in Later Childhood." *Developmental Science* 11 (3): F9–16.

Markman, Ellen M., and Gwyn F. Wachtel. 1988. "Children's Use of Mutual Exclusivity to Constrain the Meanings of Words." *Cognitive Psychology* 20 (2): 121–57.

Markson, Lori, and Paul Bloom. 1997. "Evidence Against a Dedicated System for Word Learning in Children." *Nature* 385 (6619): 813–15.

Mayor, Julien, and Kim Plunkett. 2010. "Vocabulary Spurt: Are Infants Full of Zipf?" In *Proceedings of the 32nd Annual Conference of the Cognitive Science Society*. Vol. 32.

McMurray, Bob. 2007. "Defusing the Childhood Vocabulary Explosion." *Science* 317 (5838): 631–31.

McMurray, Bob, Jessica S Horst, and Larissa K Samuelson. 2012. "Word Learning Emerges from the Interaction of Online Referent Selection and Slow Associative Learning." *Psychological Review* 119 (4): 831.

Mitchell, C., and B. McMurray. 2009. "On Leveraged Learning in Lexical Acquisition and Its Relationship to Acceleration." *Cognitive Science* 33 (8): 1503–23.

Mollica, Francis, and Steven T Piantadosi. 2017. "How Data Drive Early Word Learning: A Cross-Linguistic Waiting Time Analysis." *Open Mind* 1 (2): 67–77.

Muter, Valerie, Charles Hulme, Margaret J Snowling, and Jim Stevenson. 2004. "Phonemes, Rimes, Vocabulary, and Grammatical Skills as Foundations of Early Reading Development: Evidence from a Longitudinal Study." *Developmental Psychology* 40 (5): 665.

Peter, Michelle S, Samantha Durrant, Andrew Jessop, Amy Bidgood, Julian M Pine, and Caroline F Rowland. 2019. "Does Speed of Processing or Vocabulary Size Predict Later Language Growth in Toddlers?" *Cognitive Psychology* 115: 101238.

Pinstock, Eveline, and Satyam Antonio Schramm. 2026. "Vocabulary Trajectories in German-Speaking Children from 18 Months to Three Years: A Growth Mixture Model." *Journal of Child Language* 53 (1): 62–79. https://doi.org/10.1017/S0305000924000503.

Rasch, Georg. 1960. *Probabilistic Models for Some Intelligence and Attainment Tests*. Copenhagen, Denmark: Danish Institute for Educational Research.

Rescorla, Leslie. 2011. "Late Talkers: Do Good Predictors of Outcome Exist?" *Developmental Disabilities Research Reviews* 17 (2): 141–50.

Rowe, Meredith L. 2012. "A Longitudinal Investigation of the Role of Quantity and Quality of Child-Directed Speech in Vocabulary Development." *Child Development* 83 (5): 1762–74.

Rowe, Meredith L, and Catherine E Snow. 2020. "Analyzing Input Quality Along Three Dimensions: Interactive, Linguistic, and Conceptual." *Journal of Child Language* 47 (1).

Roy, Brandon C., Michael C Frank, Philip DeCamp, Matthew Miller, and Deb Roy. 2015. "Predicting the Birth of a Spoken Word." *Proceedings of the National Academy of Sciences* 112 (41): 12663–68.

Sanchez, Alessandro, Stephan C. Meylan, Mika Braginsky, Kyle E. MacDonald, Daniel Yurovsky, and Michael C Frank. 2019. "Childes-Db: A Flexible and Reproducible Interface to the Child Language Data Exchange System." *Behavior Research Methods* 51 (4): 1928–41.

Simonsen, Hanne Gram, Kristian Emil Kristoffersen, Dorthe Bleses, Sonja Wehberg, and Rune Nørgård Jørgensen. 2014. "The Norwegian Communicative Development Inventories: Reliability, Main Developmental Trends and Gender Differences." *First Language* 34 (1): 3–23. https://doi.org/10.1177/0142723713510997.

Singletary, Britt, Hui Jiang, Winifred Graham Wilberforce, Daniela Avelar, Kristina Strother-Garcia, and Laura M. Justice. 2025. "Examining Early Vocabulary Growth Trajectories in Late Talkers in a Low-Income Longitudinal Sample." *Infancy* 30 (4): e70036. https://doi.org/10.1111/infa.70036.

Smith, Linda B., Susan S. Jones, Barbara Landau, Lisa Gershkoff-Stowe, and Larissa Samuelson. 2002. "Object Name Learning Provides on-the-Job Training for Attention." *Psychological Science* 13 (1): 13–19. https://doi.org/10.1111/1467-9280.00403.

Tan, Alvin WM, Virginia A Marchman, and Michael C Frank. 2024. "The Role of Translation Equivalents in Bilingual Word Learning." *Developmental Science* 27 (4): e13476.

Thal, Donna J., Virginia A. Marchman, and J. Bruce Tomblin. 2013. "Late Talking Toddlers: Characterization and Prediction of Continued Delay." In *Late Talkers: Language Development, Interventions, and Outcomes*, edited by L. Rescorla and P. Dale. Baltimore, MD: Brookes Publishing.

Trueswell, John C, Tamara Nicol Medina, Alon Hafri, and Lila R Gleitman. 2013. "Propose but Verify: Fast Mapping Meets Cross-Situational Word Learning." *Cognitive Psychology* 66 (1): 126–56.

Warstadt, Alex, Aaron Mueller, Leshem Choshen, Ethan Wilcox, Chengxu Zhuang, Juan Ciro, Rafael Mosquera, et al. 2023. "Findings of the BabyLM Challenge: Sample-Efficient Pretraining on Developmentally Plausible Corpora." In *Proceedings of the Babylm Challenge at the 27th Conference on Computational Natural Language Learning*, 1–34.

Weisleder, A., and A. Fernald. 2013. "Talking to Children Matters: Early Language Experience Strengthens Processing and Builds Vocabulary." *Psychological Science* 24 (11): 2143–52. https://doi.org/10.1177/0956797613488145.

Woodward, Amanda L, Ellen M Markman, and Colleen M Fitzsimmons. 1994. "Rapid Word Learning in 13-and 18-Month-Olds." *Developmental Psychology* 30 (4): 553.

Zettersten, Martin, Daniel Yurovsky, Tian Linger Xu, Sarp Uner, Angeline Sin Mei Tsui, Rose M Schneider, Annissa N Saleh, et al. 2023. "Peekbank: An Open, Large-Scale Repository for Developmental Eye-Tracking Data of Children's Word Recognition." *Behavior Research Methods* 55 (5): 2485–2500.

Supplemental Information

Comparison with Prior Models

```
## Relation to prior accumulator models  [draft for promotion - MCF to review]

Our model is one member of a family of accumulator models of early
vocabulary, and the family is easiest to compare in a single set of
coordinates. Writing the latent ability of child $i$ on word $j$ at
age $t$ as

$$\eta_{ij}(t) = \log r_i + \log \alpha_i + \kappa_i \log t - \delta_j,$$

every prior model in this tradition is a special case that fixes one or
more of three quantities: the between-child variance in efficiency
($\sigma_\alpha$), the population scaling exponent ($\kappa_{\text{pop}}$),
and - critically - the between-child variance in that exponent
($\sigma_\kappa$). Mapping each prior model into these coordinates makes
the comparison structural rather than notational.

**McMurray [-@mcmurray2007]** is a pure accumulator: every word fills at a
single unit rate ($\kappa = 1$) and children do not differ
($\sigma_\alpha = \sigma_\kappa = 0$). Its central insight is that the
*vocabulary* curve accelerates even when the underlying rate does not,
because ability sweeps through a distribution of word difficulties - the
construction reproduced in panel (A) of our schematic. Acceleration here
is a population-level property of the difficulty distribution, not a
change in the child. This corresponds to our M1.

**Mitchell & McMurray [-@mitchell2009]** add a leverage term: already-known
words speed the acquisition of new ones. They prove that this feedback
shifts *when* words are acquired but cannot manufacture acceleration that
is not already present in the difficulty distribution. In our coordinates
it moves the timing of acquisition without changing $\kappa$, and - being
non-linear in accumulated vocabulary - it is the one prior model that does
not embed cleanly in the linear predictor above.

**Hidaka [-@hidaka2013]** introduces a rate-change process in which ability
grows as $t^{D+1}$; his exponent $D$ is exactly $\kappa_{\text{pop}} - 1$
in our notation, so $D > 0$ is population-level acceleration
($\kappa_{\text{pop}} > 1$). This is our M2. The acceleration is shared
across children ($\sigma_\kappa = 0$). Hidaka relates $D$ to word
imageability - i.e. a *per-word* scaling exponent - which is an
interesting direction we do not pursue here.
```

```
**Mollica & Piantadosi [-@mollica2017]** model each word as a Gamma
waiting-time process in which acquisition occurs after roughly ten
"effective learning instances." In our coordinates this is again a pure
accumulator ($\kappa = 1$); the per-word variation they capture lives in
the Gamma shape parameter, which corresponds to a per-word discrimination
(a 2PL extension) rather than to per-child structure.

**Kachergis et al. [-@kachergis2022]** are the first to free
$\sigma_\alpha$: they cast vocabulary as a Rasch model in which each child
has their own latent ability, unifying the accumulator and item-response
traditions. This is our M3. But the scaling exponent remains shared across
children ($\sigma_\kappa = 0$), and the model is proposed as a framework
rather than fit to longitudinal individual data.

Reading down this list, the literature progressively frees parameters -
McMurray fixes all three, Hidaka frees $\kappa_{\text{pop}}$, Kachergis
frees $\sigma_\alpha$ - but every model holds $\sigma_\kappa = 0$.

| Model | $\sigma_\alpha$ | $\kappa_{\text{pop}}$ | $\sigma_\kappa$ | per-word $\delta_j$ |
|---|:--:|:--:|:--:|:--:|
| McMurray (2007)             | 0 | 1     | 0 | free |
| Mitchell & McMurray (2009) | 0 | 1     | 0 | free + leverage |
| Hidaka (2013), rate-change | 0 | $D+1$ | 0 | free |
| Mollica & Piantadosi (2017)| 0 | 1     | 0 | free |
| Kachergis et al. (2022)    | free | free | 0 | free |
| **This work (M4)**         | **free** | **free** | **free** | **free** |

Our contribution is to free $\sigma_\kappa$: children differ not only in
how efficient they are but in how steeply their learning accelerates. This
is the load-bearing addition - it is what lets the model fit the widening
fan of trajectories with age (M4 vs M3 in the main text). It is also a
falsifiable claim rather than a parameter convenience: if $\sigma_\kappa$
were near zero the model would collapse onto Kachergis et al., but the
estimated per-child slope variance is well above zero and large relative
to the efficiency variance $\sigma_\alpha$ (main text, Figure 2 and
Results).
```

Nonidentifiability in Cross-Sectional Data

Datasets Used

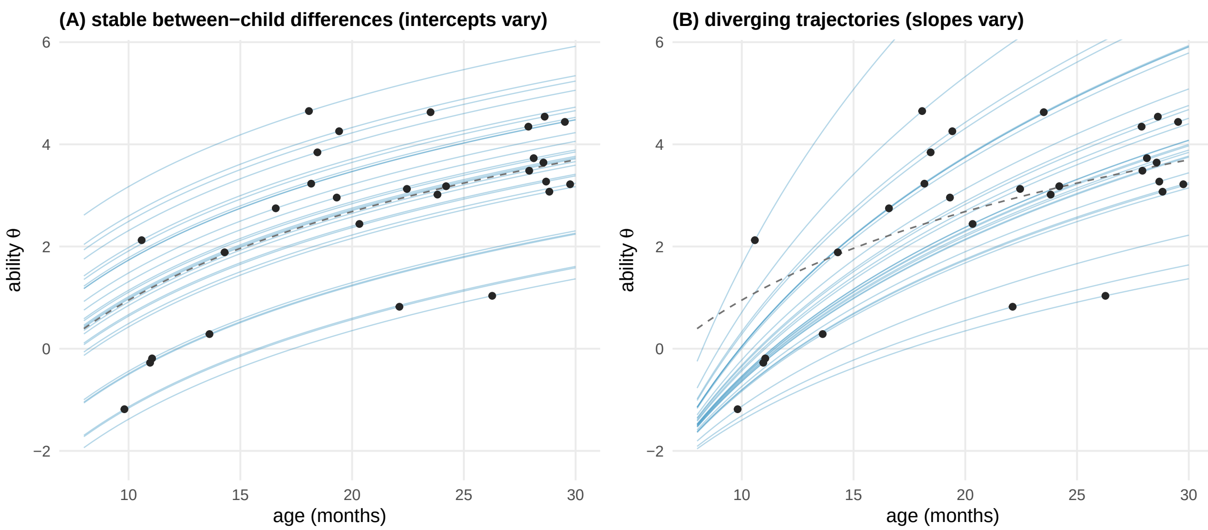

Figure 6: Non-identifiability of growth models in cross-sectional data. Panels show the same simulated data, reflecting the possibility of stable between-child intercept differences (A) vs. varying growth slopes (B).

Table 1: Characteristics of all datasets used in the paper. N and ages reflect the analysis samples (post-exclusion). Cross-sectional rows pool all contributed Wordbank datasets for a language. For AM2018 and FMW2013, which enter both the input and processing analyses with slightly different inclusion criteria, N is the union of the two subsamples.

				Age (months)						
Citation	Language	N	Admins	Mean	Min	Max	Long.	Input	Recordings	LWL sessions
Wordbank longitudinal										
Thal et al. (20XX)	English (American)	653	1,947	19.1	12	29	yes			
Smith et al. (20XX)	English (American)	314	878	21.9	16	30	yes			
Marchman et al. (20XX)	English (American)	314	714	22.6	16	30	yes			
Simonsen et al. (2014)	Norwegian	1,676	6,347	25.1	8	36	yes			
Hagihara et al. (2023)	Japanese	178	425	17.4	8	34	yes			
Input / processing										
Long et al. (2024)	English (American)	22	103	18.6	9	32	yes	headcam	5,739	
Egan-Dailey & Bergelson (2025)	English (American)	44	514	12.2	6	18	yes	LENA	525	
Adams et al. (2018)	English (American)	67	167	20.1	13	27	yes	LENA	126	440
Fernald et al. (2013)	English (American)	60	142	23.2	18	30	yes	LENA	51	93
Fernald & Marchman (2012)	English (American)	119	393	22.9	14	30	yes			419
Wordbank cross-sectional										
Frank et al. (2017)	Arabic (Saudi)	258	258	11.9	8	16				
Frank et al. (2017)	Cantonese	1,208	1,208	23.4	16	30				
Frank et al. (2017)	Catalan	1,436	1,436	19.7	8	30				
Frank et al. (2017)	Croatian	627	627	19.0	8	30				
Frank et al. (2017)	Czech	493	493	23.0	16	30				
Frank et al. (2017)	Danish	6,112	6,112	20.8	8	36				
Frank et al. (2017)	Dutch	446	446	15.6	8	31				
Frank et al. (2017)	English (American)	8,685	8,685	19.2	8	30				
Frank et al. (2017)	English (Australian)	1,493	1,493	20.6	12	30				
Frank et al. (2017)	Estonian	1,235	1,235	23.3	16	30				
Frank et al. (2017)	Finnish	234	234	16.5	9	24				
Frank et al. (2017)	French (French)	1,290	1,290	21.6	8	30				
Frank et al. (2017)	German	1,181	1,181	24.1	18	30				
Frank et al. (2017)	Hebrew	534	534	27.4	11	35				
Frank et al. (2017)	Hungarian	369	369	23.1	16	30				
Frank et al. (2017)	Italian	1,400	1,400	20.4	7	36				
Frank et al. (2017)	Japanese	846	846	20.4	8	36				
Frank et al. (2017)	Kigiriama	205	205	16.1	8	30				
Frank et al. (2017)	Korean	2,229	2,229	22.9	8	36				
Frank et al. (2017)	Latvian	683	683	22.3	8	36				
Frank et al. (2017)	Mandarin (Beijing)	1,056	1,056	23.0	16	30				
Frank et al. (2017)	Mandarin (Taiwanese)	2,654	2,654	22.0	8	36				
Frank et al. (2017)	Norwegian	7,358	7,358	23.3	8	36				

(continued)

Citation	Language	N	Admins	Age (months)			Long.	Input	Recordings	LWL sessions
				Mean	Min	Max				
Frank et al. (2017)	Portuguese (European)	4,322	4,322	19.7	8	30				
Frank et al. (2017)	Russian	1,803	1,803	21.4	8	36				
Frank et al. (2017)	Slovak	1,712	1,712	20.0	8	36				
Frank et al. (2017)	Spanish (Argentinian)	783	783	23.3	16	33				
Frank et al. (2017)	Spanish (European)	1,005	1,005	18.4	8	30				
Frank et al. (2017)	Spanish (Mexican)	1,915	1,915	19.1	8	30				
Frank et al. (2017)	Swedish	1,358	1,358	19.0	8	28				
Frank et al. (2017)	Turkish	3,537	3,537	21.4	8	36				

AIC and BIC for Longitudinal Models

Table 2

Model	Age	ΔAIC Thal	Smith	Marchman	Norwegian	Japanese
M1. accumulator	—	703,220	273,732	251,211	2,281,606	94,503
M2. + acceleration	linear	206,553	179,872	134,596	1,377,615	41,169
M2. + acceleration	log	204,930	178,471	133,877	1,348,013	39,054
M3. + efficiency var.	linear	35,856	17,418	13,752	130,625	3,850
M3. + efficiency var.	log	34,219	15,356	13,135	115,116	3,227
M4. + acceleration var.	linear	1,134	553	0	4,301	169
M4. + acceleration var.	log	0	0	30	0	0

Table 3

Model	Age	ΔBIC Thal	Smith	Marchman	Norwegian	Japanese
M1. accumulator	—	703,172	273,687	251,166	2,281,553	94,461
M2. + acceleration	linear	206,517	179,838	134,563	1,377,575	41,138
M2. + acceleration	log	204,895	178,437	133,843	1,347,973	39,023
M3. + efficiency var.	linear	35,832	17,396	13,729	130,599	3,829
M3. + efficiency var.	log	34,195	15,333	13,113	115,089	3,206
M4. + acceleration var.	linear	1,134	553	0	4,301	169
M4. + acceleration var.	log	0	0	30	0	0

Full Cross-Linguistic Model Ladder

Generating the Model-Ladder Quantile Fans

```
## How the model-ladder quantile bands are generated  [draft for promotion - MCF to review]

The percentile bands ("fans") overlaid on the empirical trajectories in the
model-ladder figures are not closed-form quantiles of a fitted distribution. For
each (dataset, model) fit we generate them by Monte-Carlo simulation: we build a
synthetic population of children and read off vocabulary percentiles at each age.

**Bootstrapping children from the fitted random effects.** For the models with a
per-child random effect (the C and D variants in both age bases) we draw
$N = 500$ synthetic children by resampling *with replacement* from the fit's
per-child BLUPs - the shrunken empirical-Bayes estimates of each observed child's
random intercept and, for the D models, random slope. The resampling uses a fixed
seed, so the bands are reproducible. The models with no per-child random effect
(the Rasch-style model A and the population-slope B models) have no child
distribution to resample and so collapse to a single median curve rather than a
fan.
```

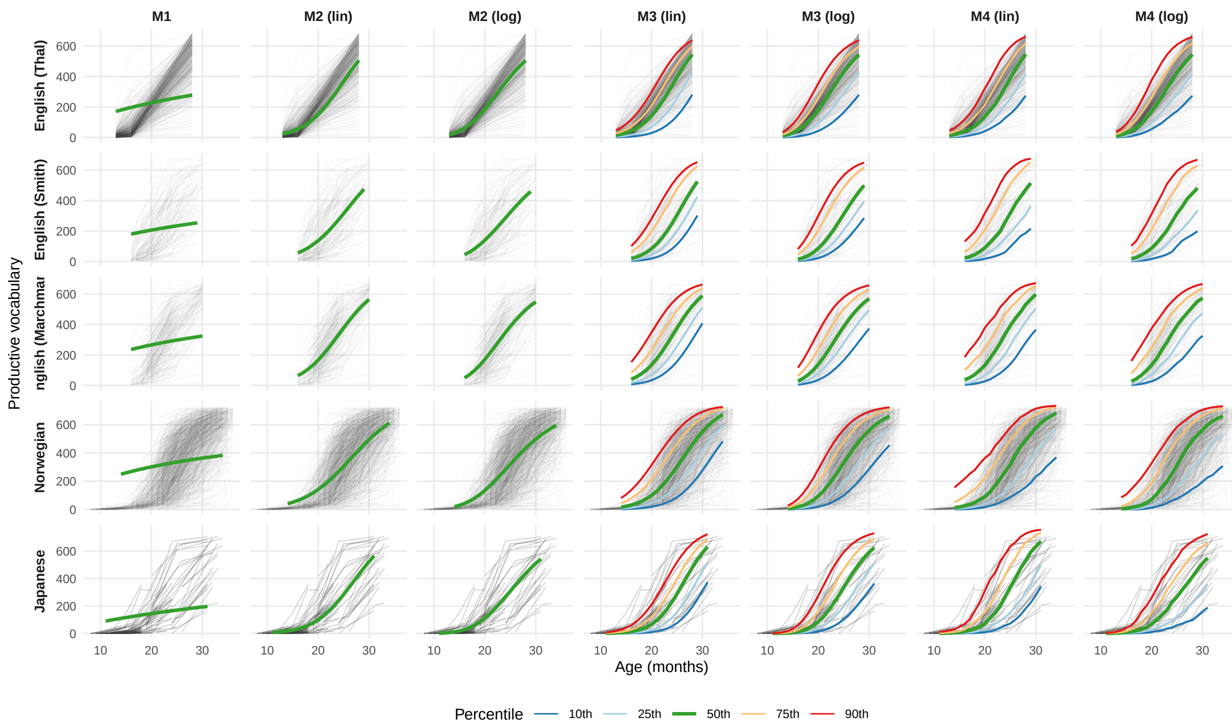

Figure 7

```
We bootstrap the realised BLUPs rather than sampling parametrically from
$\mathcal{N}(0, \hat\Sigma)$. The two agree in expectation, but parametric
sampling assumes the *unshrunken* population covariance and routinely generates
extreme intercept-slope pairs that no real child exhibited; at ages where the
data are thin this inflates the upper percentiles. Resampling the BLUPs keeps the
simulated population conditional on the children we actually observed.

**Each child's expected vocabulary curve.** Ages run on a per-dataset integer
grid spanning the 5th-95th percentile of that dataset's observed ages, so the
bands are never extrapolated beyond the data window. For a bootstrapped child $i$
with random intercept $u^{0}_i$ and slope $u^{1}_i$, the linear predictor for item
$j$ at age $t$ is
$$\eta_{ij}(t) = \beta_0 + (\beta_{\text{age}} + u^{1}_i)\, g(t) + u^{0}_i + b_j,$$
where $\beta_0$ and $\beta_{\text{age}}$ are the fixed effects, $b_j$ is item $j$'s
fitted random effect, and $g(t) = \log(t / a_0)$ for the log-age models (and for
model A, whose offset pins $\beta_{\text{age}} \equiv 1$) or $g(t) = t - a_0$ for
the linear-age models. The child's expected productive vocabulary is the sum of
per-item production probabilities over the dataset's $J$ items,
$$V_i(t) = \sum_{j=1}^{J} \operatorname{logit}^{-1}\!\big(\eta_{ij}(t)\big),$$
clamped to $[0, J]$. Item random effects are held at their fitted values - only
children are resampled - and $V_i(t)$ is an *expected* count, so each simulated
```

```
child's curve is smooth and the width of the fan reflects between-child variation
alone.

**Quantiles.** At each (dataset, model, age) the plotted bands are the 10th,
25th, 50th, 75th, and 90th percentiles of $\{V_i(t)\}_{i=1}^{500}$. The simulation
is implemented in `studies/glmer_ladder/04a_simulate.R`.
```

Population Input Range Estimation

Table 4

Source	Sample	Channel	n	tok/hr	$\pm$1 SD	tok/mo	σ_r	
Hart \& Risley	all kids pooled	mother CDS	42	1,148	546–2,410	419K	0.74	2.
Hart \& Risley	within-stratum pooled	mother CDS	42	1,148	631–2,086	419K	0.60	1.
Soderstrom \& Wittebolle	all kids pooled	all speech	12	967	622–1,502	353K	0.44	1.
Sperry	all kids pooled	adult-to-child	42	1,537	901–2,622	561K	0.53	1.
Sperry	within-stratum pooled	adult-to-child	42	1,537	974–2,425	561K	0.46	1.
Weisleder \& Fernald	all kids pooled	mother CDS	29	458	197–1,065	167K	0.84	2.
Bergelson 2018 (SEEDLingS, LENA)	US daylong, 6–7 mo	all adult (AWC)	44	1,101	767–1,580	402K	0.36	1.
Casillas 2020 (Tseltal Mayan)	Chiapas daylong	child-directed	10	720	—	263K	—	
Coffey et al. 2024 (plausible envelope)	cross-cultural bounds	CDS to all-input	—	—	4–7,232	—	—	

```
## Estimating the input-rate prior: units and conversions  [draft for promotion - MCF to review

The model represents each child's input as a log rate,
$\log r_i \sim \mathcal{N}(\mu_r, \sigma_r^2)$, measured in **adult tokens
per waking hour**. @tbl-input-rates collects published estimates of this
rate, expressed in shared units so they can be compared and so that the
implied between-child spread $\sigma_r$ is visible on one scale.

Conventions used throughout:

- **Tokens/hour to tokens/month.** We multiply by $H = 365$ waking hours
  per month ($12\text{ hr/day} \times 30.44\text{ days/mo}$), the model's
  constant.
- **$\sigma_r$ is on the log scale** - the SD of $\log(\text{tokens/hr})$
  across children within a sample. Its exponential $e^{\sigma_r}$ is the
  multiplicative factor for a one-SD shift in input (e.g.
  $\sigma_r = 0.53 \Rightarrow 1.7\times$), and the 16th-84th percentile
  spread is $e^{2\sigma_r}$.
- **Channel.** Studies count different speaker sets; we take the channel
  closest to "tokens reaching the child" - adult-to-child for Sperry,
  all-adult LENA word counts for SEEDLingS, and maternal CDS for Hart &
  Risley and Weisleder & Fernald.
- **Pooled vs. within-stratum.** For Hart & Risley and Sperry we report
  $\sigma_r$ both across all children (which mixes in SES and site
```

```
  gradients) and within stratum (subtracting each band's mean first); the
  latter is the cleaner within-population estimate.
- **Minutes to tokens.** Where a source reports minutes of speech per hour
  rather than token counts (Casillas; the Coffey bounds), we convert at
  $\approx 200$ words/min ($0.3$ s/word, the Brent-corpus CDS rate used by
  @coffey2024); those rows are accurate to roughly $\pm 20\%$.

Rows marked *computed* are recomputed here from per-dyad data; *literature*
rows are taken from the curated validation set in `input_estimation/`.
```

σ_r Sensitivity Analysis

```
## Analytic input-rate ($\sigma_r$) sensitivity  [draft for promotion - MCF to review]

The input/efficiency variance split is identified only by the external pin on
the population SD of log input rate, $\sigma_r$ - we do not observe input in the
longitudinal data. Because every reported input share therefore rests on a
quantity we assume rather than estimate, we map how the conclusion moves as
$\sigma_r$ ranges over its plausible values (panel B of @fig-io-partition).

The data identify the *total* between-child intercept variance,
$\sigma_\xi^2 = \sigma_\alpha^2 + \sigma_r^2$ (the spread of children's ability
at the reference age $a_0$), very precisely; the pin only decides how that fixed
total is *split* into an efficiency part $\sigma_\alpha^2$ and an input part
$\sigma_r^2$. Holding $\sigma_\xi^2$ at its fitted value, the input share is then
a closed form,

$$1-\pi_\alpha(\sigma_r) = \frac{\sigma_r^2}{\sigma_\xi^2},$$

which we evaluate across the plausible $\sigma_r$ range *without refitting* - the
analytic sensitivity curve.

This is first-order: it treats $\sigma_\xi^2$ as itself invariant to the pin.
Refits at off-pin values reveal a small second-order feedback - pinning a larger
$\sigma_r$ slightly *shrinks* the estimated total variance (for English,
$\sigma_\xi^2$ falls from 3.57 at $\sigma_r=0.53$ to 3.22 at $\sigma_r=0.80$). The
analytic curve therefore slightly *understates* the input share at large
$\sigma_r$ ( 18% analytic vs.  20% by refit at $\sigma_r=0.80$), an error of about
two points of $\pi_\alpha$ in the conservative direction for the size of the
input effect. The open points in panel B are these refit anchors; the gap
between point and curve is the second-order term.

We use the analytic curve, anchored by the few refits we have, rather than a
dense refit sweep: the qualitative conclusion - input accounts for a modest share
```

```
even at the top of the plausible $\sigma_r$ range - is unaffected by the
~2-point correction, whereas a full sweep of the Norwegian model alone would cost
roughly half a day of compute per pin value.
```

Computing Acceleration in LMs

```
## Comparing children and language models on one slope axis  [draft for promotion - MCF to revi

The comparison to language models in Figure 6 places children and models
on a single, literally shared axis: the slope of the acquisition sigmoid
in log-experience. Both populations have sigmoidal per-word acquisition
curves, and the slope of that sigmoid is directly comparable.

On the child side, our linear predictor makes this slope exact.
Differentiating $\eta_{ij}(t)$ with respect to $\log t$ gives $\kappa_i$:
the logit-probability that a child produces a word rises linearly in
log-age with slope equal to that child's scaling exponent. The per-child
sigmoid slope *is* $\kappa_i$, by construction.

On the model side, @chang2022 fit a four-parameter logistic to each
(model, word) pair, describing how the model's surprisal for a word falls
over training. Re-expressed as the fraction of a word's surprisal
reduction achieved so far, their fit is a sigmoid in
$\log_{10}(\text{training steps})$ with slope $1/\text{scale}_w$.
Dividing by $\ln 10$ converts this to natural-log units - the same units
as $\kappa_i$ - so the two slopes can be read off the same axis.

The result is a striking separation (Figure 6). Language models cluster
near a slope of one - the unit-accumulator boundary, where each log-unit
of additional experience yields a fixed increment of learning. Children
sit in a broad distribution roughly an order of magnitude higher. On the
shared axis, children's word learning accelerates with experience in a way
that these models' does not.

Three caveats bound this comparison. First, the variability is asymmetric:
children's slopes vary across *children*, whereas a single model is a
deterministic learner, so model slopes vary across *words* within a run
rather than across replicate learners. Second, the steep child slope is
the *per-child* quantity; a researcher fitting a single sigmoid to
cross-sectional Wordbank data per word would recover a shallower slope,
because marginalizing over the population attenuates it. Third, the models
in @chang2022 were trained on adult written text rather than
child-directed speech, so part of the gap could reflect the input
distribution rather than the learner. Models trained directly on CHILDES
```

```
(Feng et al., forthcoming) provide the matched-input test: if the gap
largely persists, it is structural; if it shrinks substantially, the input
distribution is doing more of the work than we currently attribute to it.

<!-- [BIB NEEDED] Feng et al. (CHILDES-trained GPT) - no verified entry yet;
     named in text only. Daunizeau (2017) logistic-Gaussian approximation is
     used in the derivation note but not needed for this prose. -->
```

```
## Connection to neural scaling laws  [draft for promotion - MCF to review; most speculative o

The per-word picture above connects to a more familiar aggregate
phenomenon: neural scaling laws. A model's loss falls as a smooth power
law in the amount of training data, and this descent is remarkably
regular - different runs, and different model sizes, trace nearly the same
curve (Kaplan et al., 2020; Hoffmann et al., 2022). Our per-word analysis
is the micro-level counterpart of that regularity. Each word is acquired
along a sigmoid whose slope sits near the unit-accumulator value, and the
spread of those slopes - across words and across runs - is small. The
aggregate smoothness of scaling laws and the per-word $\kappa \approx 1$
we recover are two views of the same property: language models convert
experience into knowledge as near-pure accumulators, with little
instance-to-instance variation.

Children depart from this picture on both axes at once. Their per-instance
scaling is far steeper ($\kappa \approx 10$), and they vary substantially
from one another in that exponent ($\sigma_\kappa$ well above zero). The
"data gap" between children and language models [@frank2023; @warstadt2023]
is therefore not only that children reach adult competence from orders of
magnitude less input. It is that children convert experience into knowledge
through a different scaling relationship - steeper, and far more variable
across individuals - than the one that scaling laws describe for models.

This connection is analogical rather than formal: scaling laws characterize
aggregate loss, whereas our $\kappa$ characterizes per-item acquisition,
and the two need not move together. We offer it as a framing for why the
data gap may be harder to close than raw data counts suggest.

<!-- [BIB NEEDED] Kaplan et al. (2020) and Hoffmann et al. (2022) are named
     in text but have NO verified bib entries yet - do not promote with live
     @ markers until added. -->
```

Characterizing Variation in Efficiency and Acceleration

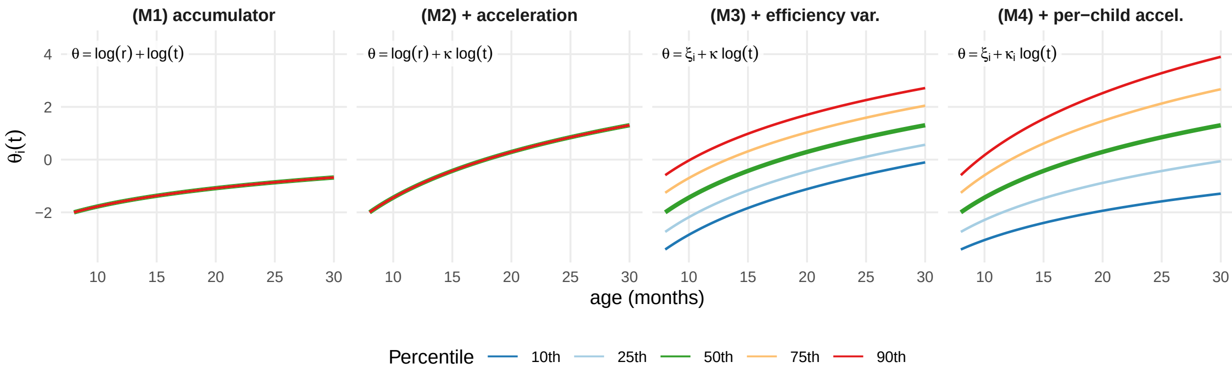

Figure 8

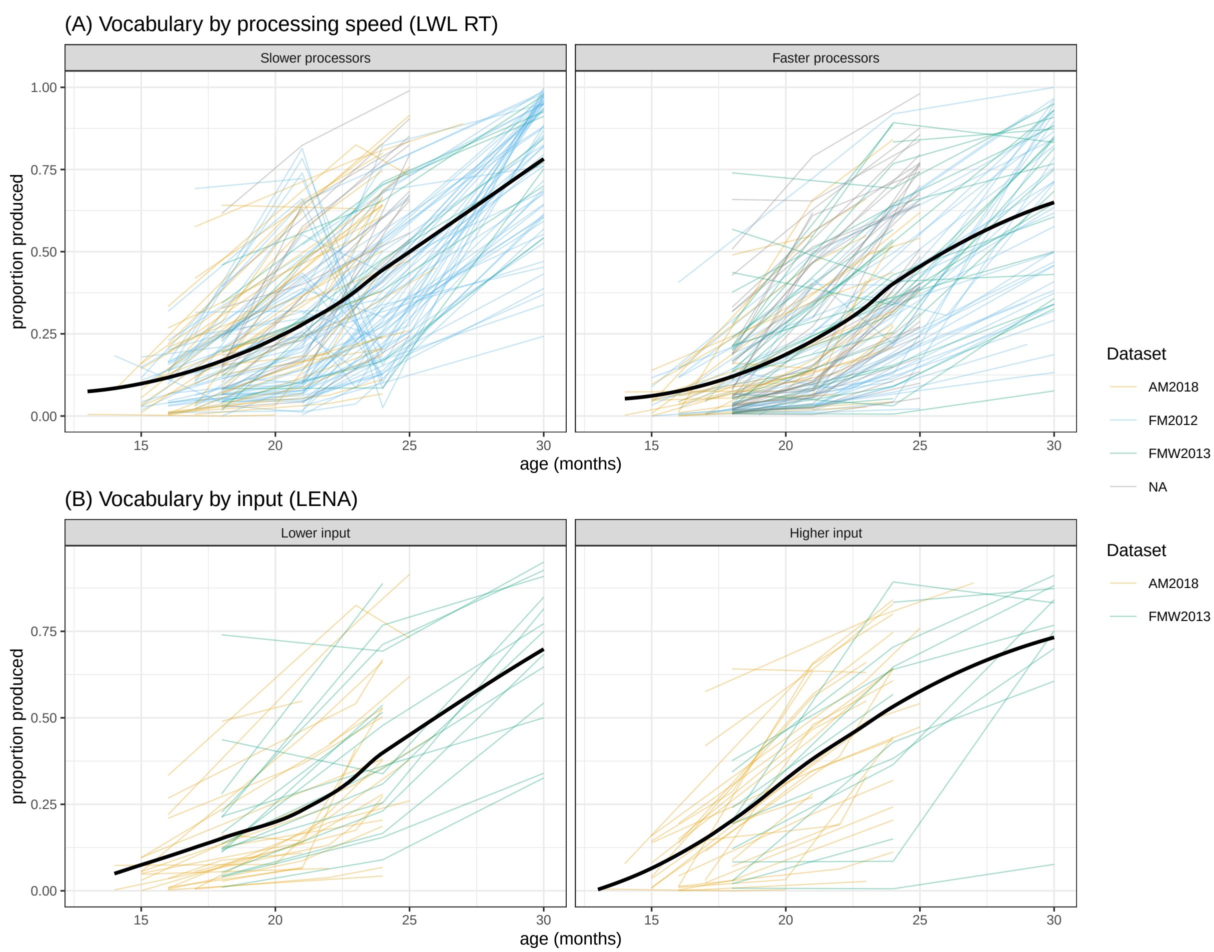

Figure 9

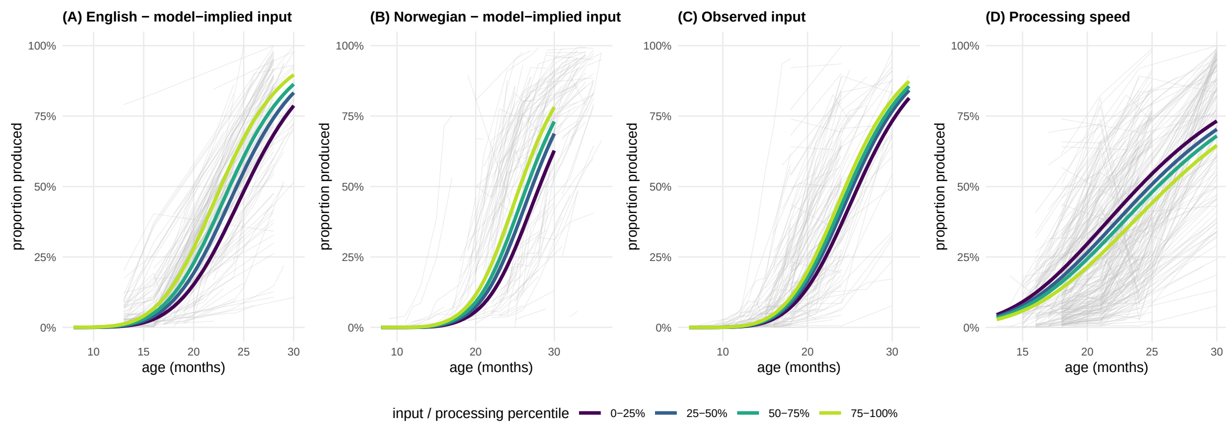

Figure 10

Table 5

Dataset	N	Efficiency (ξ)	Acceleration (ζ)
English (Thal)	651	0.96	>0.99
English (Smith)	312	0.30	0.99
English (Marchman)	313	0.99	0.99
Norwegian	1667	>0.99	0.86
Japanese	176	0.85	0.99
Pooled	3119	0.98	0.96